

Vishal Raavi

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Professional Summary

Offensive security engineer and **M.Eng. in Cybersecurity** graduate (GPA: 3.88/4.0, University of Maryland) with hands-on penetration testing experience executing full attack chains black-box reconnaissance, SQL injection exploitation, PHP webshell deployment, privilege escalation, and lateral movement across isolated target environments. 2+ years building and securing production containerized systems (**React.js, Flask, Docker**), with demonstrated ability to identify and remediate **OWASP Top 10** vulnerabilities from an attacker's perspective. Cloud security experience across **AWS** infrastructure aligned to **NIST CSF** and **CIS Benchmarks**.

Education

University of Maryland, College Park

Master of Engineering in Cybersecurity (GPA: 3.88/4.0)

Graduate Certificate in Cloud Engineering (GCEN)

College Park, MD

Jan 2024 – Dec 2025

Jan 2024 – Dec 2025

GITAM Deemed University

B.Tech, Computer Science and Engineering (GPA: 3.2/4.0)

Award: First Class with Distinction

Bangalore, India

Jun 2019 – May 2023

Certifications

- Burp Suite Certified Practitioner (BSCP) - PortSwigger (In Progress)

Technical Skills

- **Programming Languages:** Python, Java, JavaScript, C, SQL, HTML, CSS
- **Frameworks & Libraries:** React.js, Node.js, Flask, Pandas, NumPy, Tkinter, SheetJS, Plotly.js, React Flow, RESTful API
- **Penetration Testing:** Metasploit, Burp Suite, Nmap, SQLmap, Hydra, Wireshark, Weevely, Shodan, Sublist3r, arp-scan
- **Offensive Security:** SQL Injection, XSS, File Upload Exploitation, Privilege Escalation, Lateral Movement, Password Spraying, Webshell Deployment, Credential Harvesting
- **Cloud (AWS):** EC2, S3, RDS (MySQL), VPC, ALB, Auto Scaling, CloudFront, Route 53, NAT Gateway
- **Cloud Security:** WAF, IAM, KMS, ACM, CloudWatch, CloudTrail, VPC Flow Logs, Security Groups, NACLs
- **Security Frameworks:** OWASP Top 10, NIST CSF, CIS Benchmarks
- **OS & Systems Security:** SELinux, File Permission Hardening, Patch Management
- **DevOps & Tools:** Git, GitHub, GitHub Actions, Docker, Bash/Shell Scripting
- **Operating Systems:** Linux (Ubuntu, Kali Linux), Windows, macOS

Professional Experience

Center for Advanced Life Cycle Engineering (CALCE) — University of Maryland

College Park, MD

Software Developer (General Assistant)

Jan 2026 – Present

- Implementing a **cross-module data integration layer** using **REST APIs** to directly connect the reliability platform with other research modules, enabling real-time data sharing across tools and eliminating manual data entry between systems for a unified end-to-end analysis workflow.
- Built and launched the **MOSTCOOL public-facing website** (HTML5, Tailwind CSS, Vanilla JS) serving as the central hub for researchers and stakeholders to access software downloads, documentation, and team information; integrated **Google Analytics 4** to track visitor engagement and optimize content delivery.
- Developed an **automated support and distribution system** using a **token-authenticated Flask API** and GitHub Issues integration, enabling users to submit issues directly from the website and securely download research software replacing a manual process with a fully automated, access-controlled workflow.
- Conducted a full **security audit of the platform codebase**, identifying and remediating **OWASP Top 10 vulnerabilities** (XSS, Insecure Design) through input validation, CORS hardening, and DOM-safe content handling reducing software defects by **50%** and aligning the platform with industry security standards.

Software Developer (Graduate Research Assistant)

Aug 2024 – Dec 2025

- Designed and built the **MOSTCOOL Reliability Web Platform** from the ground up using **React.js, Flask** modernizing a legacy desktop application into a full-stack containerized web platform for analyzing the reliability and availability of data center cooling systems, accessible from any device with a single command; improved system response time by **50%** through optimized serialization and backend request validation.
- Built an **interactive visual modeling tool** using **React Flow** allowing researchers to construct and analyze complex system reliability diagrams through a drag-and-drop interface, eliminating manual calculations and reducing modeling time by **60%**.

- Developed the **underlying reliability calculation engine** in **Python** capable of evaluating every possible failure state across large systems to compute exact reliability and availability metrics.
- Engineered a **data export and import system** using **SheetJS** allowing researchers to save their complete work to Excel across 5 worksheets, share it across teams, and reload it with full fidelity including all diagrams, parameters, and configurations with no data loss.
- Built **cross-platform Bash/Shell deployment scripts** enabling the platform to be installed and launched consistently across **ARM64 and x86_64** architectures, removing manual setup steps and standardizing the researcher onboarding experience.

Software Developer

Apr 2024 – Aug 2024

- Built a cross-platform reliability modeling desktop tool (**Python Tkinter, NumPy**) improving modeling efficiency by **60%**; developed a responsive website (**HTML, CSS, JavaScript**) with **Google Sheets API** as a lightweight serverless backend for research data.

Security Experience

Penetration Testing & CTF Exploitation

- Executed two independent **black-box CTF** engagements from **zero knowledge to root** on isolated target VMs. Each engagement began with host discovery using **ifconfig** and **arp-scan**, followed by **Nmap** port enumeration across HTTP/80, SSH/22, and SMB/139 and 445, and manual **HTML source analysis** to map the full attack surface before any exploitation was attempted.
- Manually identified and verified a **SQL injection** entry point, then weaponized **SQLmap** to enumerate the backend database structure and extract the target table containing **CEO credentials** achieving authenticated access from a fully unauthenticated starting position and demonstrating **end-to-end database compromise** through a single web vulnerability.
- Generated a **Weevely PHP webshell** and deployed it through an unrestricted file upload endpoint to achieve **Remote Code Execution (RCE)**; pivoted through the shell to locate and exfiltrate the **admin SSH private key**, establishing persistent system access. Captured **plaintext credentials in transit** via **Wireshark** on unencrypted HTTP traffic as a parallel credential harvesting vector.
- Performed **post-exploitation enumeration** via **/etc/passwd** and mail artifacts to identify additional accounts; executed **Hydra password spray** attacks to compromise discovered users and expand foothold across the target environment.
- Achieved **root privilege escalation** by bypassing **sudo** policy restrictions; circumvented an absent **unzip** binary using a **Python-based ZIP decryption** workaround and exfiltrated final target data via **hex dump** decoded with **CyberChef**, completing the full kill chain from unauthenticated web access to root with data exfiltration.

Cloud Security Assessment — AWS VM & Vulnerability Analysis

- Authored the **Vulnerability Assessment Report** for an AWS-hosted OWASP Juice Shop platform by enumerating IAM trust relationships, auditing network boundary controls, validating EC2 hardening posture, and identifying critical misconfigurations across logging and disaster recovery domains. Findings were synthesized into a prioritized risk report aligned to **NIST CSF** and **CIS Benchmarks**.
- Performed **VM Vulnerability Assessment** on the target EC2 instance and identified a chain of exploitable misconfigurations including absent patch management, **insecure recursive file permissions** (**chmod -R 755**) exposing **.env** files and API keys to unauthorized read access, a missing host-based firewall (**iptables/nftables**), **no IDS/IPS deployment** (**Snort/Suricata**), and **SELinux operating in permissive mode** with no enforcing policy in place. Delivered a full remediation roadmap prioritized by exploitability and blast radius.
- Synthesized systemic exposure across all six assessment domains into a **security assessment conclusion** covering viable attack paths including brute force, privilege escalation, and data exfiltration, and provided actionable recommendations for least-privilege IAM restructuring, encryption at rest and in transit, IDS/IPS deployment, and CloudWatch monitoring with alerting thresholds.

Scalable and Secure E-Commerce Platform on AWS

- Designed, developed, and deployed a robust e-commerce platform leveraging a suite of **AWS** cloud services for scalability, security, and performance.
- Architected for high availability using an **Application Load Balancer (ALB)**, **Auto Scaling Groups** with **EC2** instances, and a Multi-AZ **Amazon RDS (MySQL)** database.
- Implemented multi-layer security using **AWS WAF** to filter malicious traffic (e.g., **SQL injection**), **AWS Certificate Manager (ACM)** for **HTTPS** encryption, and **AWS KMS** for data-at-rest encryption.
- Optimized global content delivery and reduced latency by configuring **Amazon CloudFront (CDN)** to cache static assets stored in **Amazon S3**.
- Established comprehensive monitoring with **Amazon CloudWatch** for application health and performance metrics, and used **AWS CloudTrail** for API action auditing.